Open Source Software Audit Report

Git – Distributed Version Control System

Student Information

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Course: Open Source Software

Selected Software: Git

Abstract

This report presents a comprehensive analysis of Git, a distributed version control system extensively used in modern software development. The study covers its background, design principles, internal structure, and role within the open-source ecosystem, particularly in Linux environments.

Along with theoretical understanding, the report includes five shell scripts that demonstrate practical knowledge of Linux commands and open-source tools. The objective of this project is to examine Git's significance, evaluate its strengths and limitations, and understand how it supports collaborative development. A comparison with proprietary tools is also included to highlight its advantages.

1. Overview of Open Source Software

Open source software refers to programs whose source code is publicly accessible. Users are allowed to view, modify, and distribute the software freely. This model differs significantly from proprietary software, where access to source code is restricted.

The open-source approach promotes collaboration, transparency, and continuous improvement. Developers from across the globe contribute to projects, making them more reliable and innovative.

Open source software is widely used in areas such as:

- Operating systems
- Web technologies
- Programming languages
- Databases
- Development frameworks

Popular examples include:

- Linux
- Git
- Apache
- Python
- Mozilla Firefox

For students and developers, open source provides an excellent opportunity to learn by exploring real-world codebases.

2. Introduction to Git

Git is a distributed version control system designed to manage changes in files over time. It is mainly used in software development but can also handle other types of projects.

Git enables users to:

Track file modifications

- Collaborate efficiently
- Maintain version history
- Work with branches
- Merge updates seamlessly

Unlike traditional systems, Git allows every user to maintain a complete copy of the repository, making it faster and more reliable.

3. Background of Git

Git was introduced in 2005 by Linus Torvalds during the development of the Linux kernel. Before Git, developers relied on a proprietary tool, which later caused licensing issues.

To overcome these limitations, Git was developed with key objectives such as:

- High efficiency
- Strong data integrity
- Distributed workflow
- Flexible branching
- Scalability

Since then, Git has become the industry standard for version control.

4. Creator of Git

Linus Torvalds, a renowned software engineer and creator of Linux, developed Git to simplify and improve large-scale project management.

His focus was on performance, simplicity, and reliability. Today, Git is used globally by millions of developers and organizations.

5. Purpose Behind Git Development

Git was designed to address several challenges faced in earlier systems, including:

- Slow operations
- Limited branching capabilities
- Dependence on centralized servers
- Poor scalability

By introducing a distributed structure, Git significantly improved development workflows.

6. Git Architecture

Git operates using a distributed model, where each user has a full copy of the repository.

It consists of three main components:

Working Directory – where files are modified

Staging Area – where changes are prepared for commit

Repository – where all versions are stored

Git uses snapshots instead of tracking individual file changes, which enhances efficiency.

7. Key Features of Git

Some of Git's important features include:

- Distributed version control
- Efficient branching and merging
- High speed performance
- Strong data integrity using hashing
- Support for large projects

8. Git Workflow

The general workflow in Git includes:

- Editing files
- Adding changes to staging
- Committing updates
- Pushing changes to remote repository

This structured approach helps maintain consistency in development.

9. Git vs Centralized Systems

Traditional systems like SVN and CVS follow a centralized approach.

- Feature Git SVN/CVS
- Architecture Distributed Centralized
- Performance Faster Slower
- Offline Work Supported Limited
- Branching Simple Complicated

Git offers more flexibility and efficiency compared to centralized systems.

10. Git vs Proprietary Tools

Git competes with proprietary tools such as BitKeeper and Perforce.

Advantages of Git:

- Free and open source
- Large community support
- High adaptability

These factors make Git a preferred choice over paid alternatives.

11. Git in Linux Environment

Git plays a crucial role in Linux-based development. It is extensively used in managing contributions to major open-source projects.

Examples include:

- Linux Kernel
- GNOME
- KDE

Most open-source repositories rely on Git for version control.

12. Installing Git on Linux

Git can be installed using system package managers:

Ubuntu:

sudo apt install git

Fedora:

sudo dnf install git

To verify installation:

git –version

13. Basic Git Commands

Common commands include:

- Initialize repository: git init
- Clone repository: git clone
- Add files: git add .
- Commit changes: git commit -m "message"
- Push changes: git push

14. Branching and Merging

Git supports multiple branches for parallel development.

Create branch: git branch feature

Switch branch: git checkout feature

Merge branch: git merge feature

This allows developers to work independently without affecting the main codebase.

15. Remote Repositories

Remote platforms enable collaboration among developers.

Popular platforms include:

GitHub

GitLab

Bitbucket

Commands:

Push: git push origin main

Pull: git pull origin main

16. Role in Open Source Development

Git simplifies collaboration by allowing:

- Forking repositories
- Submitting pull requests
- Reviewing code changes

It has become essential for open-source contribution workflows.

17. Shell Script Implementation

This project includes five scripts demonstrating Linux operations:

- System Identity Report
- Package Inspector
- Disk Audit
- Log Analyzer
- Manifesto Generator

18–22. Script Descriptions

Each script performs a specific task:

- System Identity Report: Displays system details
- Package Inspector: Checks Git installation
- Disk Auditor: Examines directory permissions
- Log Analyzer: Searches keywords in logs
- Manifesto Generator: Creates custom open-source text

23. Screenshots

```
C:\Users\visma\Desktop\OSS-AUDIT-24BCE10575>"C:\Program Files\Git\bin\bash.exe" scripts/package_inspector.sh
Git is installed
git version 2.53.0.windows.2
Git enables distributed version control

C:\Users\visma\Desktop\OSS-AUDIT-24BCE10575>
```

```
C:\Users\visma\Desktop\OSS-AUDIT-24BCE10575>"C:\Program Files\Git\bin\bash.exe" scripts/manifesto.sh
Open Source Manifesto Generator

Tool you use daily: phone
Freedom means: to live life
You would build: a world to be happy
Manifesto saved to manifesto_visma.txt
On Tue Mar 31 19:39:00 IST 2026, I reflect on open source.
I use phone regularly.
For me, freedom means to live life.
I hope to build a world to be happy and share it.

C:\Users\visma\Desktop\OSS-AUDIT-24BCE10575>
```

```
C:\Users\visma\Desktop\OSS-AUDIT-24BCE10575>"C:\Program Files\Git\bin\bash.exe" scripts/log_analyzer.sh
Analyzing log file...

Keyword 'error' found 2 times

Last matching lines:
Error occurred
Error detected again

C:\Users\visma\Desktop\OSS-AUDIT-24BCE10575>
```

```
C:\Users\visma\Desktop\OSS-AUDIT-24BCE10575>"C:\Program Files\Git\bin\bash.exe" scripts/disk_audit.sh
Directory Audit Report
----------------------

Checking Home Directory
drwxr-xr-x 1 visma 197609 0 Mar 31 19:33 /c/Users/visma

Checking C Drive
drwxr-xr-x 1 visma 197609 0 Mar 29 08:52 /c

Checking Users Directory
drwxr-xr-x 1 visma 197609 0 Oct 18 17:59 /c/Users

Checking Git Config File
-rw-r--r-- 1 visma 197609 72 Mar 31 19:10 /c/Users/visma/.gitconfig

C:\Users\visma\Desktop\OSS-AUDIT-24BCE10575>
```

```
Command Prompt
Microsoft Windows [Version 10.0.26200.8037]
(c) Microsoft Corporation. All rights reserved.

C:\Users\visma>cd "C:\Users\visma\Desktop\OSS-AUDIT-24BCE10575"

C:\Users\visma\Desktop\OSS-AUDIT-24BCE10575>"C:\Program Files\Git\bin\bash.exe" scripts/system_identity.sh
==================================
 Open Source Audit System Report
==================================
Student : VISMAY JAISWAL
Software : Git

Kernel Version : 3.6.6-1cdd4371.x86_64
Operating System : Msys
User : visma
Home Directory : /c/Users/visma
Date : Tue Mar 31 19:35:56 IST 2026

This system uses open source components under GNU GPL License

C:\Users\visma\Desktop\OSS-AUDIT-24BCE10575>
```

24. Advantages of Git

- Open source and free
- Fast and efficient
- Supports distributed work
- Ensures data security

25. Limitations of Git

- Requires initial learning effort
- Commands may seem complex for beginners

26. Applications

Git is widely used in:

- Software development
- Team collaboration
- Documentation management

27. Learning Outcomes

This project helped in understanding:

- Git concepts and workflow
- Open-source principles
- Linux shell scripting

28. Conclusion

Git is a powerful and essential tool in modern software development. Its distributed nature and efficiency make it highly suitable for collaborative environments.

This project provided both theoretical knowledge and practical exposure, enhancing understanding of open-source systems and Linux operations.

29. References

- Official Git Documentation
- Linux Documentation
- Open Source Initiative

Author

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